

2303A52416

batch=35

Task 1 – Customer Data Cleaning

Task:

Use AI to generate a Python script for cleaning a customer dataset.

Instructions:

- Handle missing values in columns (age, income, city).
- Remove duplicate records.
- Standardize text values (e.g., city names in lowercase).
- Encode categorical variables (gender, city).

Sample Code:

```
import pandas as pd
data = pd.read_csv("shopping_trends.csv")
print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset?select=shopping_trends.csv

Expected Output:

- A cleaned Pandas data frame ready for analysis.

Code

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder

# Load dataset
data = pd.read_csv("shopping_trends.csv")

# Handle missing values
data['age'].fillna(data['age'].median(), inplace=True)
data['income'].fillna(data['income'].mean(), inplace=True)
data['city'].fillna(data['city'].mode()[0], inplace=True)

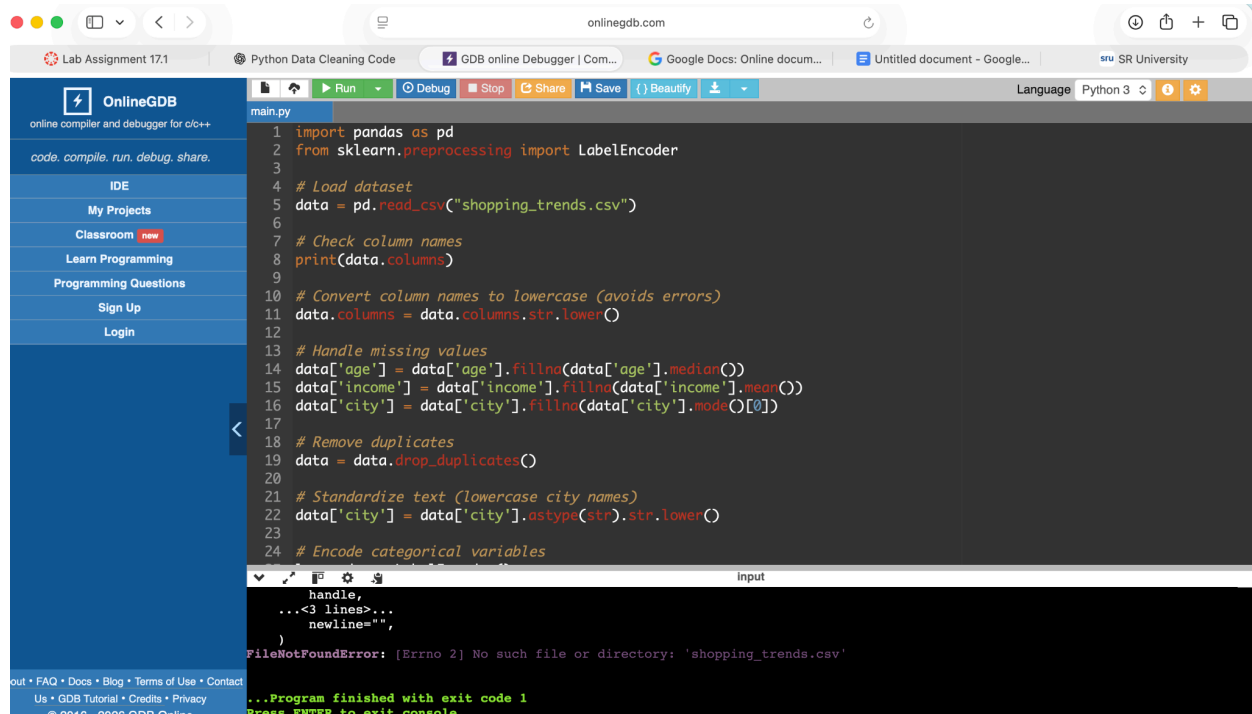
# Remove duplicates
data.drop_duplicates(inplace=True)

# Standardize text (lowercase city names)
data['city'] = data['city'].str.lower()

# Encode categorical variables
le = LabelEncoder()
```

```
data['gender'] = le.fit_transform(data['gender'])
data['city'] = le.fit_transform(data['city'])
```

```
print(data.head())
```



OnlineGDB online compiler and debugger for c/c++

code, compile, run, debug, share.

IDE

My Projects

Classroom **new**

Learn Programming

Programming Questions

Sign Up

Login

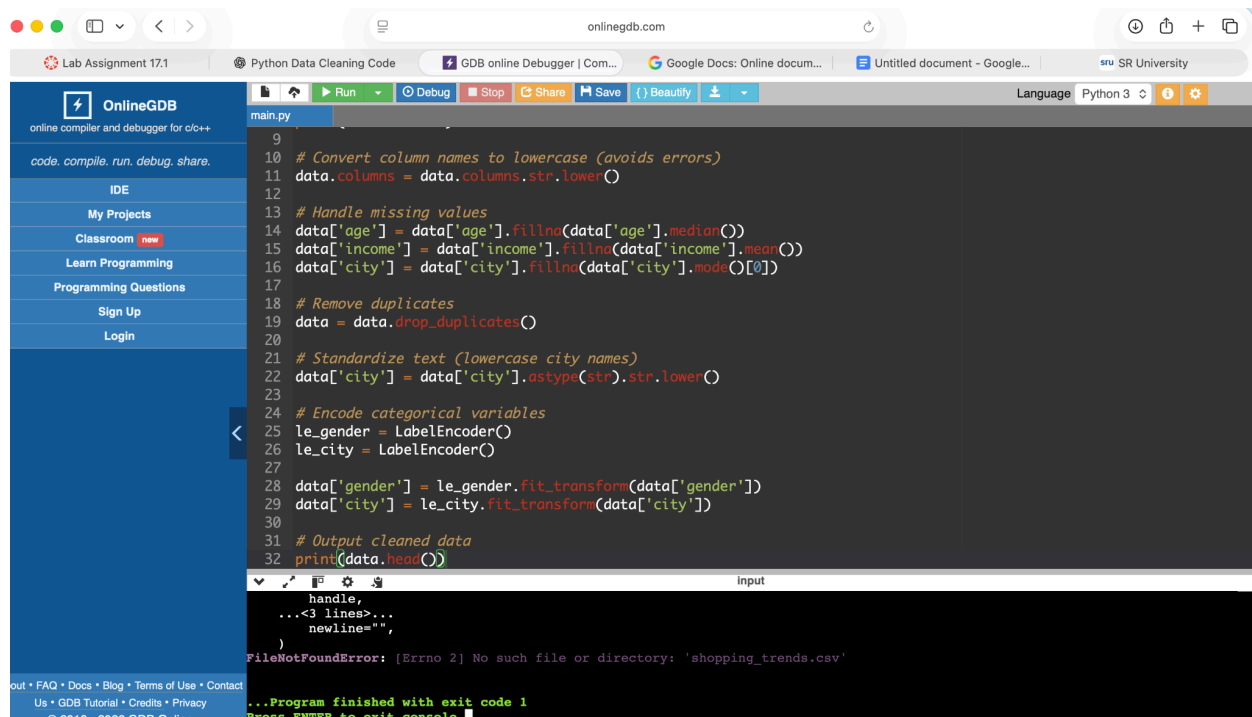
Lab Assignment 17.1 Python Data Cleaning Code GDB online Debugger | Com... Google Docs: Online docum... Untitled document - Google... SR University

main.py

```
1 import pandas as pd
2 from sklearn.preprocessing import LabelEncoder
3
4 # Load dataset
5 data = pd.read_csv("shopping_trends.csv")
6
7 # Check column names
8 print(data.columns)
9
10 # Convert column names to lowercase (avoids errors)
11 data.columns = data.columns.str.lower()
12
13 # Handle missing values
14 data['age'] = data['age'].fillna(data['age'].median())
15 data['income'] = data['income'].fillna(data['income'].mean())
16 data['city'] = data['city'].fillna(data['city'].mode()[0])
17
18 # Remove duplicates
19 data = data.drop_duplicates()
20
21 # Standardize text (lowercase city names)
22 data['city'] = data['city'].astype(str).str.lower()
23
24 # Encode categorical variables
```

Input

```
handle,
...<3 lines>...
newline="",
)
FileNotFoundError: [Errno 2] No such file or directory: 'shopping_trends.csv'
...Program finished with exit code 1
Press ENTER to exit console.
```



OnlineGDB online compiler and debugger for c/c++

code, compile, run, debug, share.

IDE

My Projects

Classroom **new**

Learn Programming

Programming Questions

Sign Up

Login

Lab Assignment 17.1 Python Data Cleaning Code GDB online Debugger | Com... Google Docs: Online docum... Untitled document - Google... SR University

main.py

```
9
10 # Convert column names to lowercase (avoids errors)
11 data.columns = data.columns.str.lower()
12
13 # Handle missing values
14 data['age'] = data['age'].fillna(data['age'].median())
15 data['income'] = data['income'].fillna(data['income'].mean())
16 data['city'] = data['city'].fillna(data['city'].mode()[0])
17
18 # Remove duplicates
19 data = data.drop_duplicates()
20
21 # Standardize text (lowercase city names)
22 data['city'] = data['city'].astype(str).str.lower()
23
24 # Encode categorical variables
25 le_gender = LabelEncoder()
26 le_city = LabelEncoder()
27
28 data['gender'] = le_gender.fit_transform(data['gender'])
29 data['city'] = le_city.fit_transform(data['city'])
30
31 # Output cleaned data
32 print(data.head())
```

Input

```
handle,
...<3 lines>...
newline="",
)
FileNotFoundError: [Errno 2] No such file or directory: 'shopping_trends.csv'
...Program finished with exit code 1
Press ENTER to exit console.
```

Task 2 – Sales Data Preprocessing

Task:

Preprocess sales transaction data using AI assistance.

Instructions:

- Convert date columns into datetime format.
- Create new features (month, year, day of week).
- Normalize sales amount column.
- Detect and handle outliers in dataset.

Sample Code:

```
import pandas as pd
data = pd.read_csv("sales_data.csv")
print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/vinothkannaece/sales-dataset?select=sales_data.csv

Expected Output:

- A transformed dataset with time-based features and normalized values.

```
4
5 # Load dataset
6 data = pd.read_csv("sales_data.csv")
7
8 # Convert date column
9 data['date'] = pd.to_datetime(data['date'])
10
11 # Create new features
12 data['month'] = data['date'].dt.month
13 data['year'] = data['date'].dt.year
14 data['day_of_week'] = data['date'].dt.dayofweek
15
16 # Normalize sales column
17 scaler = MinMaxScaler()
18 data['sales'] = scaler.fit_transform(data[['sales']])
19
20 # Detect and handle outliers using IQR
21 Q1 = data['sales'].quantile(0.25)
22 Q3 = data['sales'].quantile(0.75)
23 IQR = Q3 - Q1
24
25 data = data[(data['sales'] >= Q1 - 1.5 * IQR) & (data['sales'] <= Q3 + 1.5 * IQR)]
26
27 print(data.head())
```

FileNotFoundError: [Errno 2] No such file or directory: 'sales_data.csv'

...Program finished with exit code 1
Press ENTER to exit console.

```
1 import pandas as pd
2 from sklearn.preprocessing import MinMaxScaler
3 import numpy as np
4
5 # Load dataset
6 data = pd.read_csv("sales_data.csv")
7
8 # Convert date column
9 data['date'] = pd.to_datetime(data['date'])
10
11 # Create new features
12 data['month'] = data['date'].dt.month
13 data['year'] = data['date'].dt.year
14 data['day_of_week'] = data['date'].dt.dayofweek
15
16 # Normalize sales column
17 scaler = MinMaxScaler()
18 data['sales'] = scaler.fit_transform(data[['sales']])
19
20 # Detect and handle outliers using IQR
21 Q1 = data['sales'].quantile(0.25)
22 Q3 = data['sales'].quantile(0.75)
23 IQR = Q3 - Q1
24
```

FileNotFoundError: [Errno 2] No such file or directory: 'sales_data.csv'

...Program finished with exit code 1
Press ENTER to exit console.

Task 3 – Healthcare Data Preparation

Task:

Clean and preprocess a healthcare dataset using AI scripts.

Instructions:

- Handle missing values in blood_pressure, cholesterol.

- Encode categorical features
- Scale numerical features (min-max or standard scaling).
- Split dataset into training and testing sets.

Dataset link :

<https://www.kaggle.com/datasets/abdallaahmed77/healthcare-risk-factors-dataset>

Expected Output:

- A preprocessed dataset ready for machine learning models.

Code

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split

# Load dataset
data = pd.read_csv("healthcare.csv")

# Handle missing values
data['blood_pressure'].fillna(data['blood_pressure'].mean(), inplace=True)
data['cholesterol'].fillna(data['cholesterol'].mean(), inplace=True)

# Encode categorical features
le = LabelEncoder()
for col in data.select_dtypes(include='object').columns:
    data[col] = le.fit_transform(data[col])

# Scale numerical features
scaler = StandardScaler()
num_cols = data.select_dtypes(include=['int64', 'float64']).columns
data[num_cols] = scaler.fit_transform(data[num_cols])

# Split dataset
X = data.drop('target', axis=1)
y = data['target']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

print("Training data shape:", X_train.shape)
```

```
main.py
1 import pandas as pd
2 from sklearn.preprocessing import LabelEncoder, StandardScaler
3 from sklearn.model_selection import train_test_split
4
5 # Load dataset
6 data = pd.read_csv("healthcare.csv")
7
8 # Handle missing values
9 data['blood_pressure'].fillna(data['blood_pressure'].mean(), inplace=True)
10 data['cholesterol'].fillna(data['cholesterol'].mean(), inplace=True)
11
12 # Encode categorical features
13 le = LabelEncoder()
14 for col in data.select_dtypes(include='object').columns:
15     data[col] = le.fit_transform(data[col])
16
17 # Scale numerical features
18 scaler = StandardScaler()
19 num_cols = data.select_dtypes(include=['int64', 'float64']).columns
20 data[num_cols] = scaler.fit_transform(data[num_cols])
21
22 # Split dataset
23 X = data.drop('target', axis=1)
24 y = data['target']
25
26 handle,
27 ...<3 lines>...
28 newline="",
29 )
FileNotFoundError: [Errno 2] No such file or directory: 'healthcare.csv'
...Program finished with exit code 1
Press ENTER to exit console.
```

```
main.py
5 # Load dataset
6 data = pd.read_csv("healthcare.csv")
7
8 # Handle missing values
9 data['blood_pressure'].fillna(data['blood_pressure'].mean(), inplace=True)
10 data['cholesterol'].fillna(data['cholesterol'].mean(), inplace=True)
11
12 # Encode categorical features
13 le = LabelEncoder()
14 for col in data.select_dtypes(include='object').columns:
15     data[col] = le.fit_transform(data[col])
16
17 # Scale numerical features
18 scaler = StandardScaler()
19 num_cols = data.select_dtypes(include=['int64', 'float64']).columns
20 data[num_cols] = scaler.fit_transform(data[num_cols])
21
22 # Split dataset
23 X = data.drop('target', axis=1)
24 y = data['target']
25
26 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
27
28 print("Training data shape:", X_train.shape)
29
30 handle,
31 ...<3 lines>...
32 newline="",
33 )
FileNotFoundError: [Errno 2] No such file or directory: 'healthcare.csv'
...Program finished with exit code 1
Press ENTER to exit console.
```

Task 4 – Employee Salary Data Preprocessing

Task:

Clean and preprocess an employee salary dataset using AI scripts.

Instructions:

- Handle missing values
- Detect and remove salary outliers.

- Standardize department names (lowercase).
- Apply label encoding to job location, department.

Dataset link:

<https://www.kaggle.com/code/vishantmathur/employee-salary>

Expected Output:

- A structured dataset ready for HR analytics.

Code

The screenshot shows the OnlineGDB web interface. The left sidebar contains navigation links: OnlineGDB, code, compile, run, debug, share, IDE, My Projects, Classroom (marked as new), Learn Programming, Programming Questions, Sign Up, and Login. The main editor area displays a Python script named 'main.py' with the following code:

```

2 from sklearn.preprocessing import LabelEncoder
3
4 # Load dataset
5 data = pd.read_csv("employee_salary.csv")
6
7 # Handle missing values
8 data.fillna(method='ffill', inplace=True)
9
10 # Remove salary outliers using IQR
11 Q1 = data['salary'].quantile(0.25)
12 Q3 = data['salary'].quantile(0.75)
13 IQR = Q3 - Q1
14
15 data = data[(data['salary'] >= Q1 - 1.5 * IQR) & (data['salary'] <= Q3 + 1.5 * IQR)]
16
17 # Standardize department names
18 data['department'] = data['department'].str.lower()
19
20 # Label encoding
21 le = LabelEncoder()
22 data['job_location'] = le.fit_transform(data['job_location'])
23 data['department'] = le.fit_transform(data['department'])
24
25 print(data.head())

```

Below the code editor is an 'Input' field and an 'Output' field. The output field displays the following error message:

```

FileNotFoundError: [Errno 2] No such file or directory: 'employee_salary.csv'

```

At the bottom of the output field, it says: "...Program finished with exit code 1" and "Press ENTER to exit console."